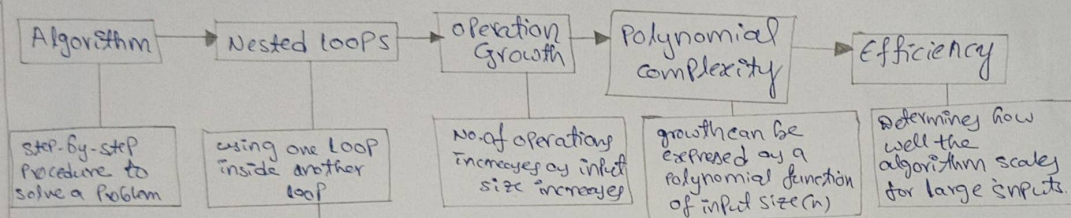


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Q7) consider the concept Map:- Algorithm $\rightarrow$ Nested loops $\rightarrow$ operation Growth $\rightarrow$ Polynomial complexity $\rightarrow$ efficiency. explain how multiple nested loops increase the no. of operations.



WHY NESTED LOOPS INCREASE OPERATIONS?

each additional loop multiplies the number of iterations.
If a loop runs n times and another loop runs n times inside it, total operations become $n \times n = n^2$

EXAMPLES (EXTENDING THE MAP)

$O(n^2)$

Two nested loops

```

for i = 1 to n
  for j = 1 to n
    // operation
  
```

total operation $= n \times n = n^2$

$O(n^3)$

Three nested loops

```

for i = 1 to n
  for j = 1 to n
    for k = 1 to n
      // operation
    
```

total operations $= n \times n \times n = n^3$

WHY POLYNOMIAL COMPLEXITY AFFECTS SCALABILITY?

$\rightarrow$ As n grows, $n^2, n^3, \dots$ grow very fast.
 $\rightarrow$ small increase in input size causes huge increase in operations.
 $\rightarrow$ Requires more time and resource.
 $\rightarrow$ Not suitable for very large inputs.

HOW THIS MAP HELPS IDENTIFY INEFFICIENT ALGORITHM?

$\rightarrow$ Highlights the impact of nested loops.
 $\rightarrow$ helps Predict operation growth early.
 $\rightarrow$ Identifies algorithm with identify higher polynomial orders (e.g., n^3)
 $\rightarrow$ guides us to optimize by reducing or eliminating nested loops.

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Q78] consider the concept map:- Algorithm $\rightarrow$ Input Reduction $\rightarrow$ Binary search logarithmic complexity & efficiency. Explain how reducing input size at each steps leads to logarithmic growth.

